EduBridge Enterprise Solution Architecture & Master Prompt

Executive Summary

EduBridge is a self-hosted AI-powered learning ecosystem combining Duolingo-style language learning, E2Language-style exam preparation, AI tutor capabilities, academic learning, career guidance, parent and teacher portals, gamification, portfolio building and offline-first learning.

Final Master Prompt

Build EduBridge using Next.js 15, NestJS, PostgreSQL 17, Redis, MinIO, Meilisearch, Keycloak, Ollama, NGINX, Docker, Prometheus and Grafana. Use Phi-4 Mini, Qwen 3 8B, DeepSeek 8B and Llama 3.3 8B through Ollama. Implement AI Tutor, Language Learning, IELTS/PTE preparation, Parent Portal, Teacher Portal, Career Navigator, Skill-to-Income Hub, Study Planner and Portfolio Builder.

UI/UX Architecture

Mobile-first responsive design. Child, Teen, Adult, Parent and Admin dashboards. Accessibility compliance, dark mode, PWA, offline sync, gamification, streaks, badges, leaderboards and adaptive learning.

Frontend Architecture

Next.js, React, TypeScript, TailwindCSS, ShadCN UI, Zustand, TanStack Query, React Hook Form, Zod, Service Workers and IndexedDB.

Backend Architecture

NestJS modular monolith, REST APIs, Swagger, JWT, RBAC, audit logs, background jobs, Redis cache and event-driven modules.

AI Architecture

Ollama hosted locally. Phi = grammar, Qwen = tutor, DeepSeek = reasoning and mathematics, Llama = career advisor. Whisper for speech-to-text and Piper for text-to-speech.

MVP Roadmap

Phase1 Foundation, Phase2 AI Tutor, Phase3 Language Learning, Phase4 IELTS/PTE, Phase5 Parent Portal, Phase6 Career Foundation.

Database Design

Core domains: Security, Users, Profiles, Learning, Vocabulary, Exams, Gamification, AI Conversations, Study Planner, Careers, Notifications, Files and Audit Logs.

ER Diagram

ROLES->USERS->PROFILES; COURSES->MODULES->LESSONS->LESSON_PROGRESS; EXAMS->EXAM_SECTIONS->EXAM_QUESTIONS->EXAM_ATTEMPTS->EXAM_ANSWERS; AI_CONVERSATIONS->AI_MESSAGES.

API Contracts

Base URL /api/v1. Auth, Users, Courses, Lessons, Vocabulary, Exams, AI, Study Plans, Career, Parent, Teacher and Admin modules.

Deployment Architecture

NGINX, Frontend, Backend, PostgreSQL, Redis, MinIO, Meilisearch, Ollama, Keycloak, Prometheus, Grafana, Backup service.

NGINX

TLS termination, reverse proxy, gzip, caching, security headers and rate limiting.

CI/CD

GitHub Actions: lint, tests, build, security scan, docker image build and deployment.

Security

RBAC, OAuth2, JWT, Keycloak, encryption, monitoring, backups and audit logging.

Production Docker Services

postgres, redis, minio, meilisearch, ollama, keycloak, backend, frontend, nginx, prometheus, grafana, node-exporter, cadvisor, postgres-backup.

Database Tables

Detailed implementation specifications for Database Tables.

Foreign Keys & Indexes

Detailed implementation specifications for Foreign Keys & Indexes.

Production Docker Compose Requirements

Detailed implementation specifications for Production Docker Compose Requirements.

Health Checks

Detailed implementation specifications for Health Checks.

Resource Limits

Detailed implementation specifications for Resource Limits.

Monitoring

Detailed implementation specifications for Monitoring.

Backup & Recovery

Detailed implementation specifications for Backup & Recovery.

Infrastructure Sizing

Detailed implementation specifications for Infrastructure Sizing.

Scaling Strategy

Detailed implementation specifications for Scaling Strategy.